

Cambridge Lower Secondary Checkpoint

CANDIDATE
NAME

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CENTRE
NUMBER

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CANDIDATE
NUMBER

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MATHEMATICS

0862/01

Paper 1

April 2025

1 hour

You must answer on the question paper.

You will need: Geometrical instruments
Tracing paper (optional)

INSTRUCTIONS

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- You should show all your working in the booklet.
- You are **not** allowed to use a calculator.

INFORMATION

- The total mark for this paper is 50.
- The number of marks for each question or part question is shown in brackets [].

This document has **16** pages.

- 1 The table gives some information about sequence A and sequence B .

	First five terms	Next term	n th term
Sequence A	1, 4, 9, 16, 25	36	n^2
Sequence B	2, 5, 10, 17, 26		

- (a) Complete the table with the next term in sequence B .

[1]

- (b) Complete the table with the n th term of sequence B .

[1]

- 2 Tick (✓) to show if each piece of data is quantitative or qualitative.

	Quantitative	Qualitative
The name of the country where a person is born.	☐	☐
The number of books a person owns.	☐	☐

[1]

- 3 Draw a ring around the most appropriate unit for measuring the volume of water in a raindrop.

microlitre

tonne

milligram

micrometre

[1]

- 4 Mia buys some sweets.

Toffees cost 9 cents each.

Chocolates cost 11 cents each.

Mia buys t toffees and c chocolates.

The total cost of the sweets she buys is 129 cents.

Draw a ring around the equation that represents this information.

$$129 = (9 - t) + (11 - c)$$

$$129 = 9t + 11c$$

$$129 = \frac{9}{t} + \frac{11}{c}$$

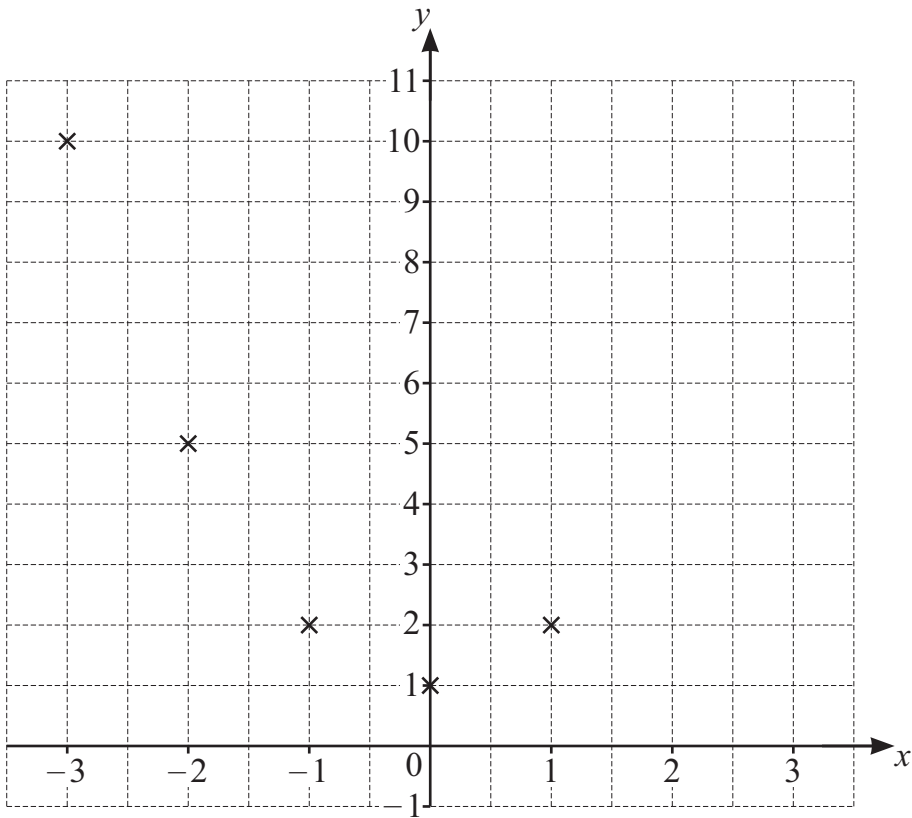
$$129 = t + 9 + c + 11$$

[1]

- 5 Make m the subject of the formula $h = m^3$

$$m = \dots\dots\dots [1]$$

6 Chen starts to draw the graph of $y = x^2 + 1$



Complete the graph of $y = x^2 + 1$ for values of x between -3 and 3
You may use the table to help you.

x	-3	-2	-1	0	1	2	3
y	10	5	2	1	2		

[2]

- 7 Safia draws triangle P on a coordinate grid.
 She rotates triangle P by 90° anticlockwise about centre $(0, -3)$ to make triangle Q.
 She then translates triangle Q by $\begin{pmatrix} 3 \\ 7 \end{pmatrix}$ to make triangle R.

Tick (✓) to show if each pair of triangles is congruent, not congruent or you cannot tell.

	Congruent	Not congruent	Cannot tell
Triangle P and triangle Q	☐	☐	☐
Triangle P and triangle R	☐	☐	☐

[1]

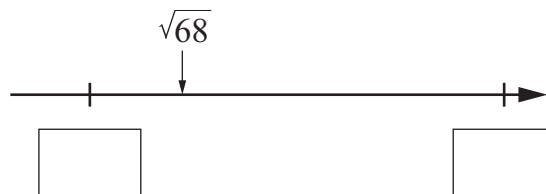
- 8 Work out.

$$\left(1 + \frac{1}{5}\right) \div \left(2 - \frac{1}{2}\right)$$

Give your answer as a fraction in its simplest form.

..... [3]

- 9 The diagram shows a number line.



Complete the boxes with the two closest **whole** numbers to $\sqrt{68}$

[1]

- 10** Lily wants to find the average height of the 1000 students in her school. The students in her school are aged between 4 and 18 years old. Here is her plan to collect a sample of students.

Choose one class of 18-year-old students in her school.
Choose 5 students randomly from this class.

Give **two** different reasons why her sampling method is biased.

- 1
.....
2
.....

[2]

- 11** Here is an equation.

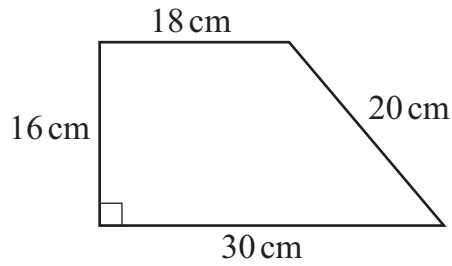
$$a^b = 16$$

Complete the table with different pairs of values for a and b .

First pair	$a =$ and $b =$
Second pair	$a =$ and $b =$
Third pair	$a =$ and $b =$

[3]

12 Here is a trapezium.



NOT TO
SCALE

Some students try to calculate the area of the trapezium in different ways.

(a) For each student, tick (✓) to show if their calculation is correct or **not** correct.

First student	$16 + 18 + 20 + 30$	Correct ☐	Not correct ☐
Second student	$16 \times 18 + \frac{1}{2} \times 12 \times 16$	Correct ☐	Not correct ☐

[1]

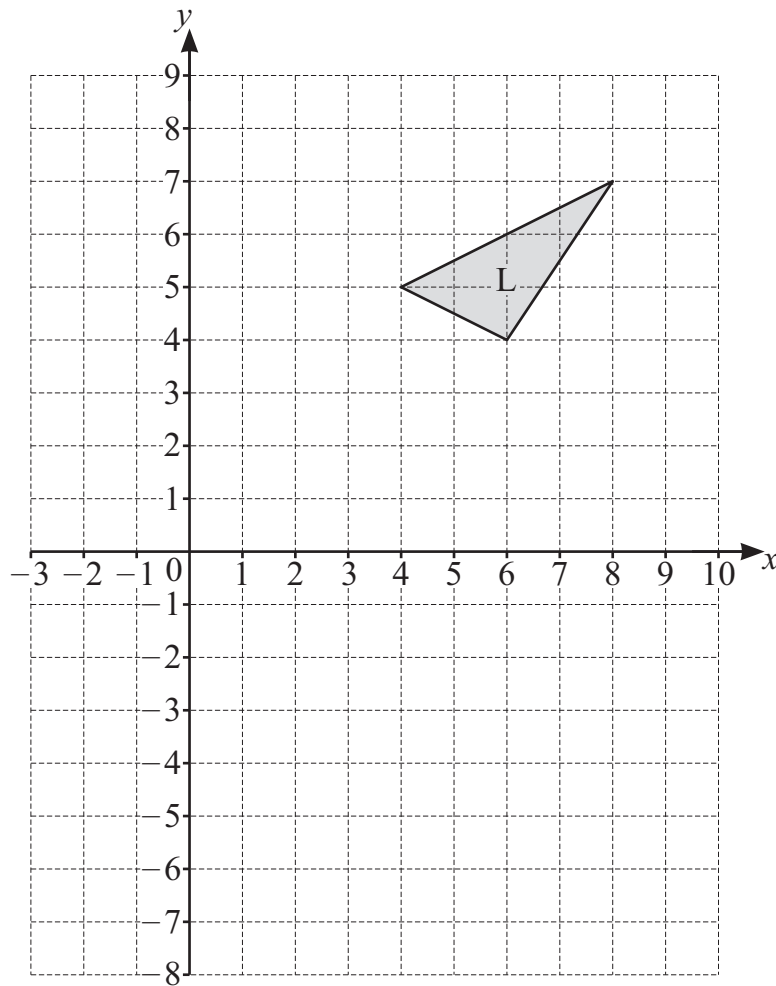
(b) A third student tries to calculate the area of the trapezium.
Here is their calculation.

$$\frac{1}{2} \times (18 + 30) \times 20$$

Draw a ring around the number that is **not** correct in the third student's calculation.

[1]

13 The diagram shows triangle L drawn on a grid.



Triangle L is reflected in the line $y = 2$ to make triangle M.

Triangle M is then translated by $\begin{pmatrix} 0 \\ -3 \end{pmatrix}$ to give triangle N.

- (a) Draw the position of **triangle N** on the grid.
You do **not** need to rub out your triangle M.

[2]

- (b) Complete the sentence with the correct words from the box.

a translation

a reflection

a rotation

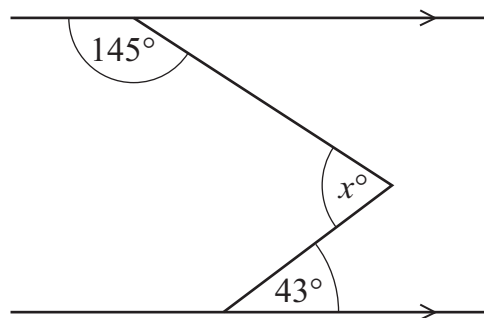
an enlargement

The **single** transformation that maps triangle L onto triangle N

is

[1]

- 14 The diagram shows two parallel lines and two other straight lines.



NOT TO
SCALE

Find the value of x .

$x =$ [2]

- 15 Work out the value of $\left(\frac{55 - 8x}{x}\right)^4$ when $x = 5$

..... [2]

16 Solve the simultaneous equations.

$$\begin{aligned}x - 3y &= 21 \\ 2x + y &= 7\end{aligned}$$

$$x = \dots\dots\dots$$

$$y = \dots\dots\dots$$

[3]

17 Work out.

$$6^2 + \left(5.3 - \frac{3}{10}\right)^2$$

$$\dots\dots\dots [2]$$

18 Naomi thinks of two integers.

n is a **negative** integer.

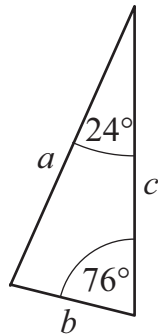
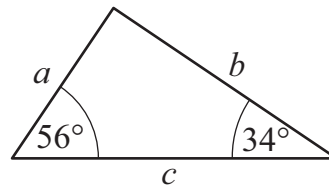
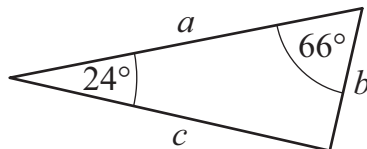
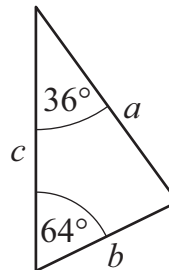
p is a **positive** integer.

For Naomi's integers $n^2 > p^2$

Write down a possible value of n and a possible value of p for Naomi's integers.

$n =$ and $p =$ [1]

19 Tick (✓) to show the triangle where $a^2 + b^2 = c^2$


☐

☐

☐

☐

[1]

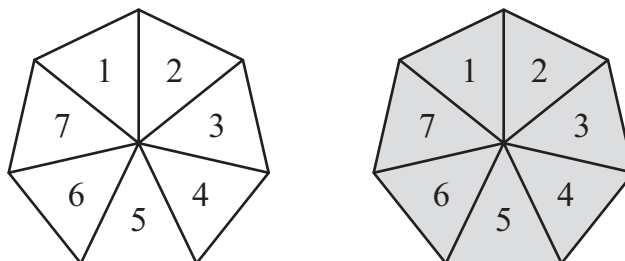
- 20** Oliver has two boxes of pencils.
The first box contains $(5x + 11)$ pencils.
The second box contains $(7x - 3)$ pencils.
The number of pencils in each box is the same.

Oliver says, 'There are more than 40 pencils in the first box.'

Show that Oliver is correct.

[3]

- 21** Yuri has two fair 7-sided spinners each numbered 1 to 7. One spinner is white and the other spinner is grey.



- (a)** Yuri spins the white spinner once and the grey spinner once. Here are two events, A and B.

Event A Spinning an even number on the white spinner.

Event B Spinning a 3 on the grey spinner.

Draw a ring around the best word or words to describe the pair of events, A and B.

dependent mutually exclusive independent complementary

[1]

- (b)** Yuri spins the white spinner 140 times.

Calculate how many times Yuri should expect to spin a number less than 3

..... [1]

- (c)** Yuri spins the white spinner once and the grey spinner once.

Calculate the probability that both spinners show an odd number.

..... [2]

22 Rajiv says,



All fractions with an odd denominator will be equivalent to a recurring decimal, for example $\frac{2}{9} = 0.\dot{2}$

Find a fraction with an odd denominator to show that Rajiv is **not** correct.

..... [1]

23 Find the value of x and the value of y in these equations.

$$(19^3)^5 \times 19^{-4} = 19^x$$

$x =$

$$7 \times 7^y \div 7^{-4} = 7^{13}$$

$y =$

[2]

- 24 A rectangle has an area of 2 m^2 .
When the rectangle is enlarged the image has an area of 72 m^2 .

Find the value of the scale factor of the enlargement.

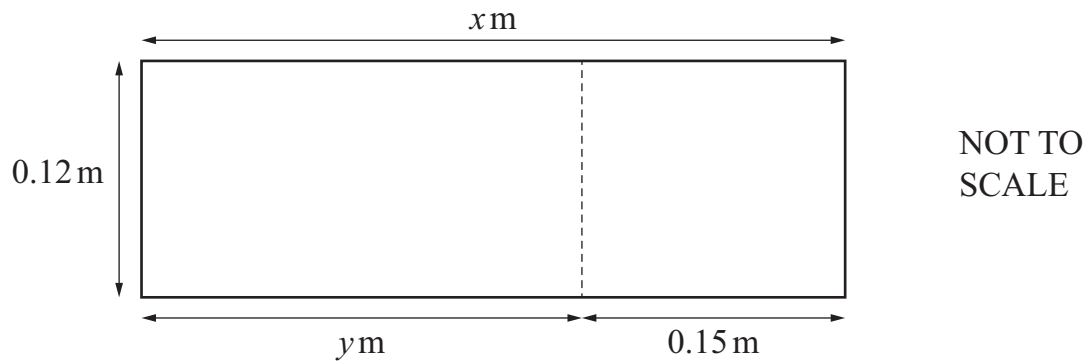
..... [1]

- 25 Simplify.

$$\frac{6tw + 9t}{3t}$$

..... [2]

26 The diagram shows a rectangle.



The total area of the rectangle is 0.084 m^2 .

Work out the value of x and the value of y .

$x =$
 $y =$

[3]

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